Geophysical Journal International

Geophys. J. Int. (2010) **180,** 147–157 doi: 10.1111/j.1365-246X.2009.04406.x

Magnetic history of Early and Middle Ordovician sedimentary sequence, northern Estonia

J. Plado,[1] U. Preeden,[1] L. J. Pesonen,[2] S. Mertanen[3] and V. Puura[1]

[1] *Department of Geology, University of Tartu, Vanemuise* 46, 51014 *Tartu, Estonia. E-mail: juri.plado@ut.ee*
[2] *Division of Geophysics, University of Helsinki,* PO Box 64, FIN-00014 *Helsinki, Finland*
[3] *Geological Survey of Finland, Betonimiehenkuja* 4, FIN-02151 *Espoo, Finland*

Accepted 2009 September 29. Received 2009 September 21; in original form 2009 January 23

SUMMARY

Alternating field and thermal demagnetization of lime- and dolostones from the Lower and Middle Ordovician (Floian to Darriwilian stages) subhorizontally bedded sequences in NW and NE Estonia reveal two characteristic magnetization components (named P and S). The intermediate-coercivity (demagnetized at 30–60 mT, up to 300–350 °C) reversed polarity component P (mean of Floian Stage: $D_{ref} = 147.8 \pm 10.8°$, $I_{ref} = 65.8 \pm 5.4°$; combined mean of Dapingian and Darriwilian stages: $D_{ref} = 166.0 \pm 8.4°$, $I_{ref} = 56.1 \pm 6.5°$) is regarded as the primary remanence of early diagenetic (chemical) origin. On the Baltica's apparent polar wander path (APWP), the palaeopoles (Floian: $Plat = 25.0$ °N, $Plon = 50.8$ °E, $K = 52.7$, $A_{95} = 7.2°$; Dapingian and Darriwilian: $Plat = 11.4$ °N, $Plon = 39.1$ °E, $K = 33.8$, $A_{95} = 6.7°$) are placed on the Lower and Middle Ordovician segment. The poles indicate that Estonia was located at southerly latitudes, decreasing with time (Floian: ∼48 °S; Dapingian and Darriwilian: ∼37 °S), when the remanence was acquired. A high-coercivity and high-unblocking-temperature component S (mean of samples: $D_{ref} = 33.7 \pm 6.3°$, $I_{ref} = 51.9 \pm 5.7°$) that is regarded as a secondary remanence has both normal and reversed polarities. On the European APWP, its palaeopole ($Plat = 52.5$ °N, $Plon = 157.9$ °E, $K = 38.9$, $A_{95} = 5.3°$) gives middle to late Permian age. According to mineralogical (SEM and optical microscopy) and rock magnetic (three-component induced remnant magnetization) studies, component P is carried by magnetite (coexisting with glauconite) and component S by haematite. Magnetite is of chemical origin, formed in the course of early diagenesis and/or dolomitization. During the Permian continental period haematite, the carrier of component S, was likely precipitated from oxidizing meteoric fluids in the already existing or simultaneously formed pore space between the dolomite crystals.

Key words: Palaeomagnetism applied to geologic processes; Remagnetization; Europe.

1 INTRODUCTION

During the last decades, since the first report of the Ordovician palaeopole of Baltica (Claesson 1978), several palaeomagnetic studies have been carried out to specify the late Precambrian to early Palaeozoic palaeogeographic history of Baltica (for the most recent overview see, e.g. Cocks & Torsvik 2005). Nowadays, the Ordovician and post-Ordovician APW path of Baltica is well determined, especially, when compared to the Ediacaran to Cambrian paths (Torsvik & Rehnström 2001; Hartz & Torsvik 2002; Popov *et al.* 2002). Lack of sedimentary sequences for the last 300–400 Ma and, therefore, lack of material suitable for direct geological and geochemical studies, amplifies the importance of palaeomagnetic and rock magnetic studies in specifying the late Palaeozoic to Cenozoic geological history of the region. By applying palaeomagnetic and rock magnetic methods, we have studied the Estonian Early to Middle Ordovician sequences. The main aim of the studies has been to (i) identify remanence components acquired at different geological stages, (ii) specify the sedimentational and post-sedimentational history of the area and (iii) fill the regional cap between the palaeomagnetic studies of Ordovician sedimentary rocks in Scandinavia (e.g. Torsvik & Trench 1991a,b; Perroud *et al.* 1992) and St Petersburg area (Smethurst *et al.* 1998).

The Lower Ordovician in Estonia is characterized by predominantly terrigenous deposits, almost uncompacted quartzose sandstones and argillite shales. These are overlain by grey-coloured shallow-water carbonate deposits formed in a large epicontinental sea that occupied a wide area of Baltica during the latest Early Ordovician up to the end of Ordovician. The whole Ordovician sequence is up to about 180-m thick, almost complete, and stratigraphically well studied by means of different fossil groups (e.g. Männil & Meidla 1994). The whole sequence slightly tilts southwards. The dip of tilting, however, is only a fraction of degree, which makes the standard palaeomagnetic tilt test impossible to apply. Because

© 2009 The Authors
Journal compilation © 2009 RAS

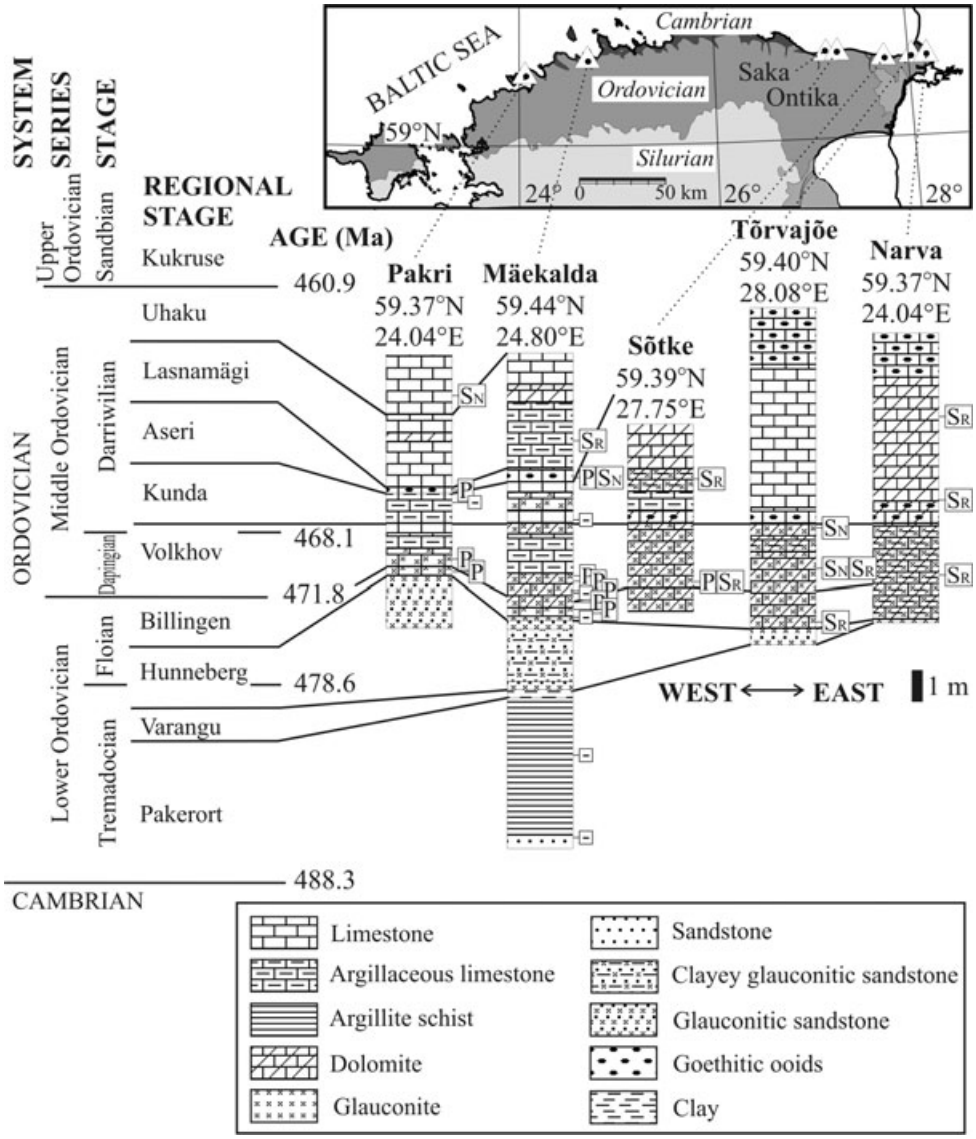


Figure 1. Stratigraphic logs for the lower and middle Ordovician in northern Estonia (see text for references). Squares with letters at the right sides of logs denote sampling locations. P = primary component; S = secondary component; indexes N and R denote normal and reversed polarities; – = samples lack characteristic components. Outcrop area of systems in central and northern Estonia is shown, including location of the sampling sites by light triangles. The logs for Saka and Ontika outcrops are not given as they did not reveal any stable palaeomagnetic results.

of tilting, younger sediments cover the sequence towards south and therefore the Ordovician outcrops occur only at the northern margin of Estonia (Fig. 1), along the Baltic Klint. As the sequence has never been buried deeply (see Kirsimäe *et al.* 1999; Kirsimäe & Jørgensen 2000) nor tectonized, the rocks are unmetamorphosed. Fig. 1 shows the global and regional Ordovician stratigraphy with stratigraphic logs and sampling locations in N Estonia.

The Tremadocian (Lower Ordovician) Pakerort regional stage that was accessed only in Mäekalda (Fig. 1) consists of two different lithotypes—the lower *Obolus* sandstone and upper *Dictyonema* shale. The weakly cemented light sandstone contains interbeds of dark argillite and its uppermost few tens of centimetres are strongly pyritizied. The upper shale is represented by dark horizontally laminated graptolite argillite with relatively high content of organic matter. The 40-cm-thick Varangu Stage in the Mäekalda section is composed of greenish-grey silty clay overlain by loose clayey glauconitic sandstone of Hunneberg (late Tremadocian—early Floian) Stage. In NW Estonia, the Billingen Stage (late Floian) is transitional from lower sandstones to upper limestones as it consists of glauconitic sandstones that are replaced upwards by calcareous sandstones and glauconitic lime- or dolostones. In the NW region, the stage is mostly composed of glauconitic dolostones. The Volkhov regional stage belongs to Dapingian and earliest Darriwilian. It is distributed all over Estonia with a thickness of few metres at the northern coast area and is composed of dolomitizied/partly dolomitizied glauconitic limestones with argillaceous interbeds. The central member of this stage may lack the glauconite compound. Glauconitic and sandy lime- and dolostones represent the Kunda Stage. The basal part of the Kunda section often contains abundant ferriferous ooids, developed around skeletal particles or glauconite grains (Mägi 1984). Thickness of the stage decreases towards NW Estonia. The Aseri Stage, 0.1–5 m thick in northern Estonia, consists of bioclastic limestones with unevenly distributed ferriferous ooids. Limestones of Lasnamägi and Uhaku stages were

sampled in Mäekalda and Pakri sites only, respectively (Fig. 1). The limestones are bioclastic, partly argillaceous and dolomitizied and consist of marly intercalations. These stages are seen only in the western part of the Baltic Klint. The Kunda to Uhaku regional stages belong to the global Darriwilian Stage. The western sequence is only sporadically dolomitizied, whereas the rate of dolomitization in the uppermost part of the carbonate sequence is higher in NE Estonia (e.g. Kiipli 1983). Stratigraphical and palaeontological details of the sampled sections are outlined by Einasto & Mens (1996) (Pakri), Viira *et al.* (2001) and Mens *et al.* (1989) (Mäekalda), Kaljo *et al.* (1986) (Saka), Mägi (1984) (Ontika), Orviku (1940) (Sõtke), Orviku (1960) (Tõrvajõe), Viira *et al.* (2006) (Narva).

In places, the Ordovician sequence is dolomitizied. Based on crystallographic and compositional studies of the Palaeozoic sequence in S Estonia, with a special focus on cyclic appearance and regional distribution of dolostones, Teedumäe *et al.* (2001, 2006) concluded that dolostones belong to shallow-water environment and represent the uppermost sediments of the regressive phases of sedimentary cycles whereas sediments of deeper environments remain not dolomitizied. The occurrence of dolostones between the undolomitizied rocks limits the time span of dolomitization to early organic matter diagenesis, meaning that dolomite likely formed close to the water–sediment interface in the sulphate reduction zone. However, Kiipli (1983) has suggested a much later and purely chemical origin based on trace element studies. The author supposes that dolomitization took place in an intermixture zone of fresh groundwater and percolating sea water from the Devonian palaeobasin. Also, hydrothermal origin of dolomitization has been proposed for the fracture zone-related sulphide-bearing dolostone bodies (Vaher *et al.* 1962). Occasionally in sequences from NE Estonia, the sediments are coloured red and this impregnation occurs unrelated to stratigraphic layering as seen in certain outcrops of different stages. As will be shown in this study, we relate the impregnation to a late Palaeozoic (middle and late Permian) event caused by circulation of meteoric fluids.

2 SAMPLES AND METHODS

Eighty-eight oriented block samples were collected from seven outcrops (Fig. 1) along the northern coast of Estonia during 1996–2002. All the outcrops, except Mäekalda, are of natural origin located along the Baltic Klint. The Mäekalda section is opened in a road-cut along the highway connecting two districts in Tallinn. In addition, the foundation trench of the present Art Museum close to the Mäekalda section was sampled. Usually, two to three samples were taken from one level. Stratigraphically, Pakerort to Uhaku regional stages were sampled (corresponding to Floian to Darriwilian global stages, see Fig. 1), concentrating on the lowermost part of the carbonate sequence as it is best accessible. Standard cylindrical specimens (height 2.1 cm, diameter 2.5 cm) were prepared from the cores. All the petrophysical (apparent density and magnetic susceptibility), rock- and palaeomagnetic studies were carried out at the laboratories of the Geological Survey of Finland and University of Helsinki.

Mineralogy (Fig. 2) of a number of samples was studied in thin sections by optical microscope at the Department of Geology, University of Tartu. Scanning electron microscope (SEM) studies and electron probe microanalyser (EPMA) studies were performed at the Geolaboratory of the Geological Survey of Finland and at the Institute of Electron Optics, in Oulu University, respectively. To identify the carriers of remanence, three-component induced remnant magnetization (IRM; with maximum available magnetizing field of 1.5 T) and subsequent stepwise thermal demagnetization in air (Lowrie 1990; Fig. 3) was carried out for 14 specimens. Thermomagnetic analysis [magnetic susceptibility (χ) vs. temperature], with heating in argon gas, was performed for a few specimens.

For palaeomagnetic studies, most of the specimens were stepwise demagnetized with alternating field (AF) up to 0.16 T. In several cases, the demagnetization was stopped earlier, if the intensity of NRM decreased below the level of instrumental noise ($\sim$0.03 mA m^{-1}). After each step, the intensity and direction of NRM were measured using a superconducting (SQUID) magnetometer. About one fifth of specimens were thermally demagnetized up to 680 °C or to a thermal step of 250–500 °C if drastic increase in magnetic susceptibility or remanence, reflecting mineralogical changes (oxidization) due to heating, took place. Individual NRM measurements were subjected to a joint analysis of stereographic plots, demagnetization decay curves, orthogonal demagnetization diagrams (Zijderveld 1967) and principal component analysis (Kirschvink 1980), to obtain the characteristic remanence directions. Fisher (1953) statistics was used to calculate mean remanence directions. Samples from Saka and Ontika outcrops did not reveal any stable remanence components and therefore those results are not shown in the following.

3 RESULTS

3.1 Physical properties

The densities of carbonate rocks are mostly between 2500 and 2700 kg m^{-3} (Table 1). The lowest values are measured from Aseri Stage limestones, which can be attributed to higher clay content and porosity (Jõeleht & Kukkonen 2002). Densities of the Pakerort schists and sandstones differ from the carbonates significantly (see Table 1). Generally, excluding the Aseri Stage limestones, the densities of limestones and dolostones increase with susceptibility. The magnetic susceptibility and intensity of NRM are generally higher in rocks including glauconite (Billingen and Volkhov stages) or ferriferous ooids (Aseri Stage). Regionally, slightly higher values are measured from the north-eastern sites. The magnetic susceptibility of the pyritizied sandstone of the Pakerort Stage is very low (mean $= 4 \times 10^{-6}$ SI), which with low intensity of NRM produces a relatively high Koenigsberger (Q) value. Intensities of the NRM are typically low for all marine sediments. Only the carbonate specimens with NRM > 0.5 mA m^{-1} and Q-value >0.2 (Table 1) carry the palaeomagnetic signal.

3.2 Mineralogy

To evaluate the grain size, position and amount of opaque minerals, standard petrographic microscopic analyses were carried out (Figs 2a, c and e). The SEM and EPMA studies of thin sections revealed iron oxides and substantial amounts of pyrite (Figs 2b, d and f). Mineralogy of carbonates differs significantly between the studied locations and stages. No primary iron oxides were identified in the limestones from NW Estonia by microscopy. However, the glauconitic limestones (Fig. 2a) were found to include rare single opaque crystals of pyrite that in some crystals is partly replaced by iron oxides (Fig. 2b). Few glauconite grains show brownish outer shells composed of iron oxides. In the samples from NE Estonia that are red or show reddish patches, fillings of iron oxide and Fe–Ti oxide (most likely secondary haematite and/or titanohaematite) were

© 2009 The Authors, *GJI*, **180,** 147–157
Journal compilation © 2009 RAS

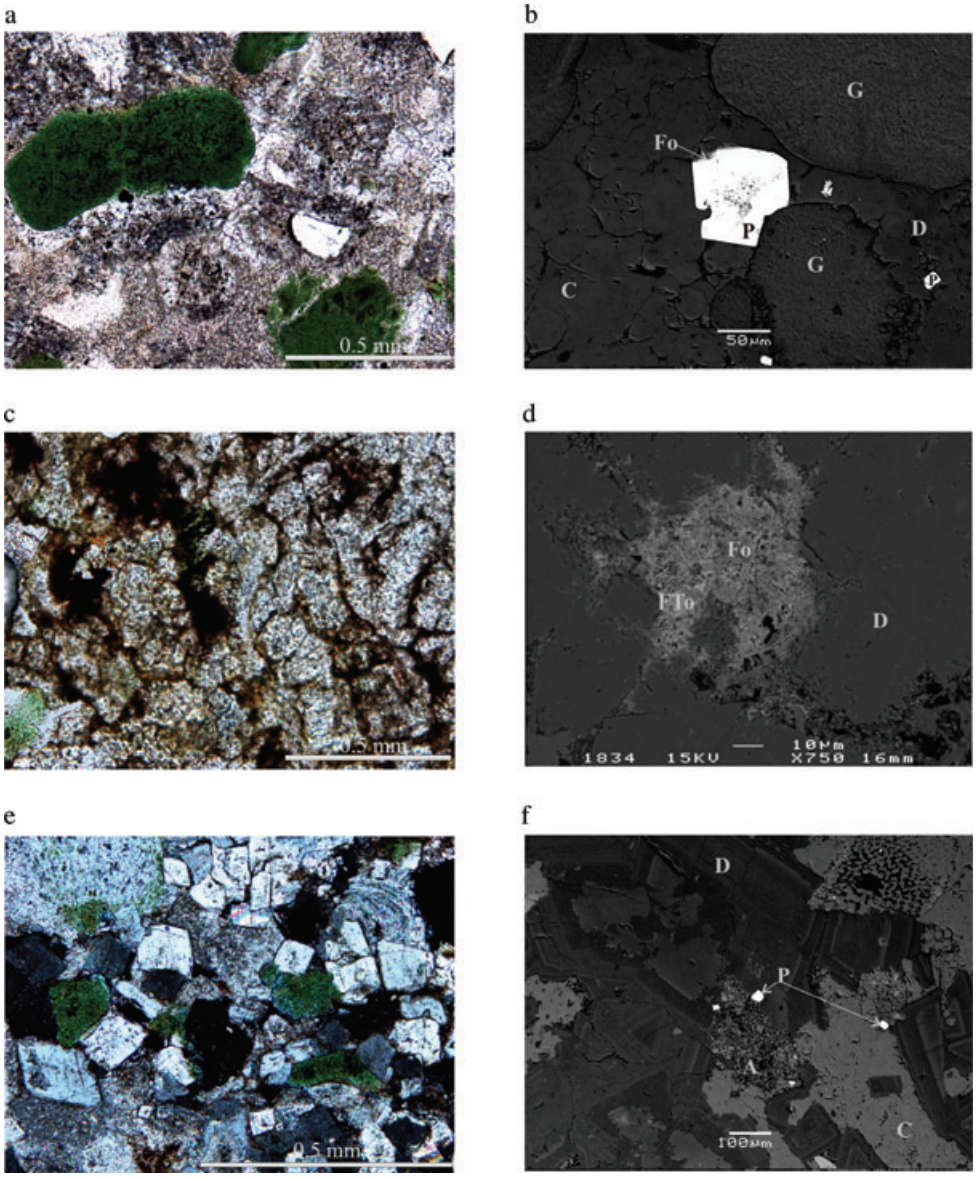


Figure 2. Microscopic (a, c and e) and backscattered scanning electron microscopic (b, d and f) images of studied carbonate thin sections. (a) Large ovoidal glauconite grains (green) in glauconitic limestone from Mäekalda (plane polarized light; sample AA12); (b) SEM backscattered image from the same sample, where in the centre of the photo the alteration of pyrite grain is seen as oxidation of the edges; (c) subhedral to anhedral dolomite crystals from Tõrvajõe site (sample TB1) and the reddish-brown material filling the voids. According to SEM studies, the reddish-brown material is composed of iron oxide and iron–titanium oxide (photo d); (e) coarsely crystalline, euhedral replacement dolomite in glauconitic limestone from Sõtke (crossed nicols; sample SE2), as with electron microscopic studies it is observed that mostly unaltered large dolomite crystals are associated with several pyrite framboid spherules (photo f). Symbols: Fo = iron oxide; P = pyrite; D = dolomite; C = calcite; G = glauconite; FTo = iron–titanium oxide; A = apatite.

found around subhedral to anhedral dolomite crystals (Figs 2c and d). In grey samples, planar euhedral dolomite crystals and glauconite grains occur in calcite matrix (Fig. 2e). They contain pyrite, but not iron oxides (Fig. 2f). The iron oxides, associated with both authigenic minerals and as pyrite replacement, were found only in association with cracks, veins, grain boundaries and other pathways for fluids. There exists an inverse correlation between observable amounts of iron oxides and sulphides, for example in red-coloured samples there are always iron oxides, but no or very little pyrite.

The IRM acquisition curves for all studied samples reveal that the samples are not totally saturated at 1.5 T. Two different behaviours were observed according to the location and stratigraphic stage (Fig. 3a). *First*, the reddish samples from NE Estonia have smooth IRM acquisition curves, indicating the dominance of high-coercivity magnetic minerals (haematite/goethite). *Second*, samples from the grey sections in NW Estonia acquire IRM rapidly up to ∼0.2 T. Then the curves continue to climb gently with somewhat different slopes between 0.2 and 0.7 T. These curves indicate that low- to medium-coercivity magnetic minerals (magnetite) are dominant, however because saturation is still not achieved at 1.5 T, small amounts of high-coercivity minerals are present as well.

Thermal demagnetizations of 3-D IRM for the same specimens as in the IRM acquisition curves show that haematite in a wide range of grain sizes is dominant in the dolostones of NE Estonia (Figs 3b–f). Minerals with lower Curie temperatures may be present in small amounts as seen from the different slopes of the

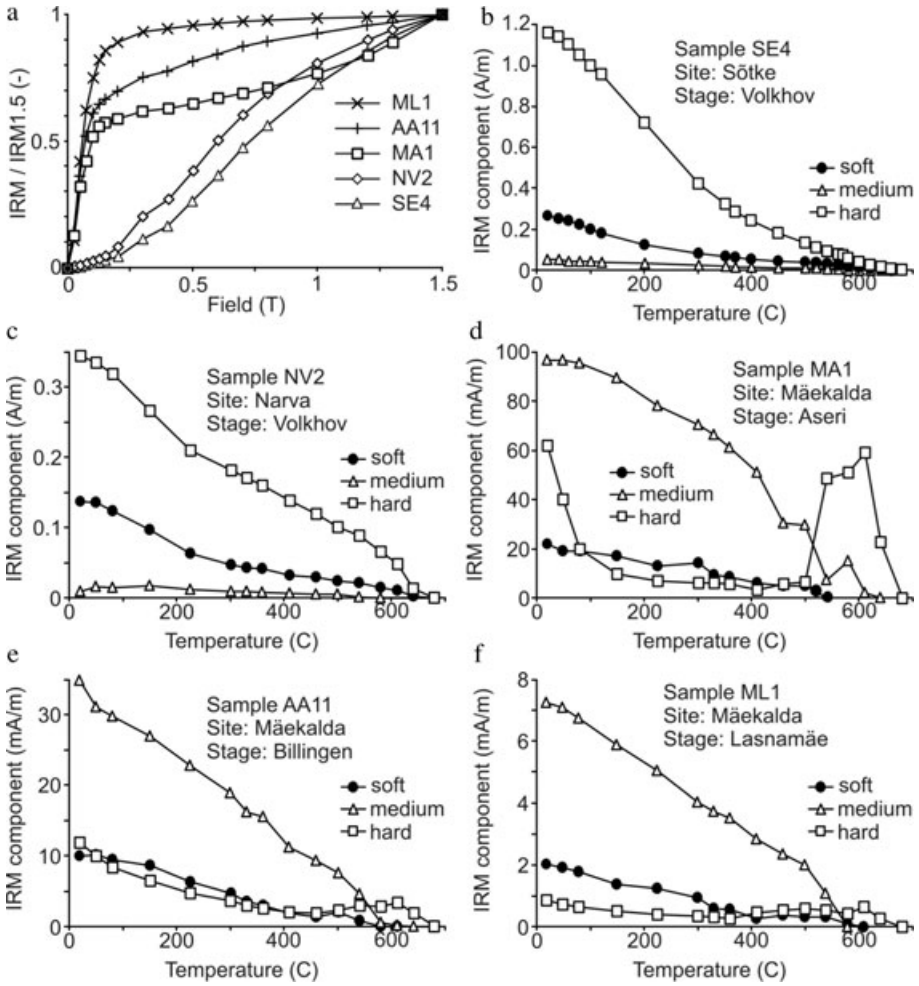


Figure 3. (a) Progressive acquisition of IRM and (b–f) thermal demagnetization of a three-component IRM produced by magnetizing the sample in 1500 mT along its *z*-axis, followed by 400 mT along the *y*-axis, and finally 120 mT along the *x*-axis.

Table 1. Average values and standard deviations of physical properties of carbonates from the different regional stages, calculated from site means.

Regional stage	Lithology	PS	N (n)	ρ_A (kg m^{-3})	χ ($\times 10^{-6}$ SI)	NRM (mA m^{-1})	Q
Uhaku	Bioclastic limestone	S_N	1/3/18	2661	42	0.06	0.15
Lasnamägi	Bioclastic limestone, occasionally argillaceous	S_R	2/6/36	2642	49	0.05	0.03
Aseri	Limestone with ferriferous or phosphatic ooids	$P + S_N$	3/11/54	2510 ± 14	91 ± 26	1.74 ± 0.97	0.50 ± 0.23
Kunda	Lime- and dolostone, occasionally with glauconite and ferriferous ooids	$S_N + S_R$	6/16/97	2650 ± 65	115 ± 76	0.09 ± 0.03	0.09 ± 0.15
Volkhov	Predominantly glauconitic lime- and dolostone	$P + S_N + S_R$	6/29/208	2661 ± 50	192 ± 71	1.08 ± 1.18	0.23 ± 0.26
Billingen	Glauconitic lime- and dolostone	$P + S_R$	4/14/87	2663 ± 24	167 ± 63	0.80 ± 1.07	0.24 ± 0.40
	Glauconitic sandstone	–	1/3/18	2519	336	0.21	0.02
Pakerort	Argillite schist	–	1/3/11	1948	78	0.14	0.04
	Pyritizied sandstone	–	1/3/22	3198	4	0.18	1.01

PS = palaeomagnetic signature, where P (S) denotes primary (secondary) remanence component. Subindex N (R) denotes normal (reversed)-polarity of the characteristic remanence; *B*/*N (n)* = number of sites (samples); ρ_A = apparent density; χ = magnetic susceptibility (volume normalized); NRM = intensity of the natural remanent magnetization; Q = Koenigsberger ratio. To link the regional stages with global ones, see Fig. 1.

demagnetization curves but the data do not allow proper identification. Most likely, some magnetite is present in sample NV2 (Narva site, Volkhov Stage), according to the decay of the intermediate coercivity component at 580 °C. Samples from different stages of Mäekalda in the western study area (Figs 3d–f) show the dominance of intermediate-coercivity component, related to magnetite. Thermal demagnetization shows that goethite constitutes the ferriferous ooids in Aseri Stage of the Mäekalda section (Fig. 3d). The sample shows an abrupt increase in the intensity of the hard-coercivity component between 500 °C and 530 °C (increase starts somewhat earlier

© 2009 The Authors, *GJI*, **180,** 147–157
Journal compilation © 2009 RAS

Table 2. Remanence directions and corresponding virtual geomagnetic pole positions as calculated from sample means.

Component (*site*)	Polarity	*B*/*N*/*n*	$D \pm \Delta D$ (°)/$D_{ref} \pm \Delta D_{ref}$ (°)	$I \pm \Delta I$ (°)/$I_{ref} \pm \Delta I_{ref}$ (°)	*Plat*	*Plon*	*K*	A_{95}
P_F *(Pakri)*	R	1/*3/16	131.2 ± 16.6	64.7 ± 8.8	23.2	58.3	120.6	11.3
P_F *(Mäekalda)*	R	1/*6/20	145.6 ± 16.1	69.4 ± 6.6	25.8	47.0	49.5	9.6
P_F	**R**	**2/*9/36**	**147.8 ± 10.8**	**65.8 ± 5.4**	**25.0**	**50.8**	**52.7**	**7.2**
P_{DD} *(Pakri)*	R	1/*6/26	150.0 ± 10.3	63.1 ± 5.9	16.9	45.9	84.5	7.3
P_{DD} *(Mäekalda)*	R	1/*5/17	171.5 ± 19,9	60.0 ± 13.0	10.6	31.3	27.5	14.9
P_{DD} *(Sõtke)*	R	1/*4/30	157.2 ± 16.2	51.9 ± 14.9	3.8	38.6	46.9	13.6
P_{DD}	**R**	**3/*15/77**	**166.0 ± 8.4**	**56.1 ± 6.5**	**11.4**	**39.1**	**33.8**	**6.7**
S_N	N	3/*5/25	**27.4 ± 16.2**	**57.7 ± 11.7**	59.6	162.3	37.6	12.6
S_R	R	4/*15/50	**215.6 ± 6.9**	**−49.8 ± 6.6**	49.8	156.8	43.3	5.9
S_{N+R}	**N+R**	**4/*20/75**	**33.7 ± 6.3**	**51.9 ± 5.7**	**52.2**	**157.9**	**38.9**	**5.3**

Comp. = magnetic component; *B*/*N*/*n* = number of sites/samples/specimens revealing the component (the level of statistics is given by asterisks); *D* and *I* = declination and inclination of the mean magnetization with confidence limits (ΔD and ΔI); D_{ref} and I_{ref} (numbers given in bold) = reference declination and inclination of the mean magnetization with confidence limits (ΔD_{ref} and ΔI_{ref}), calculated for a common reference locality at Kajaani (64.1 °N, 27.7 °E) using equations published by Butler (1992); *Plat* and *Plon* = the latitude and longitude of the pole; *K* = Fisher's (1953) precision parameter; A_{95} = the radius of a cone of 95% confidence about the mean pole.

at about 400 °C) that is likely due to dehydration of goethite into haematite. We suggest that pyrrhotite or maghaemite, responsible for the failure in thermal demagnetization process at higher temperatures, is present in most of the samples. It is demonstrated by a sharper decay of the soft-coercivity component between 300 °C and 400 °C (Fig. 3f), and oxidation-caused increase of the hard component at higher temperatures (Figs 3d–f) during the thermal demagnetization of a 3-D IRM.

3.3 Palaeomagnetic behaviour

Two characteristic remanent magnetization components (hereafter called P and S) were identified (Table 2). The components were separated by directional groupings and by coercivity and thermal spectra (Figs 3 and 4). Component P is present in glauconitic lime- and dolostones of Billingen, Volkhov and Aseri stages, preferably in NW Estonia. The other characteristic component (S) is best represented in the reddish haematite-bearing samples in NE Estonia, independently of stratigraphic stages, but it appears occasionally in the upper part of the Pakri sequence as well. The two components differ according to their stability against AF demagnetization. Component P is of intermediate coercivity, removed at fields below 60 mT. Component S is more resistant to AF, as it is not removed at 160 mT, the maximum field applied (Figs 4e and g). Only reversed polarity P directions were identified, but component S carries both polarities S_N and S_R. They have antipodal symmetry, and the sample means just pass the reversal test (category C; McFadden & McElhinney 1990). During the course of thermal demagnetization the components are observable up to 300–400 °C, after which an abrupt increase in magnetic susceptibility and/or intensity of the NRM occurs, indicating mineralogical changes. Comparing the demagnetization results with the magnetic mineralogy (Fig. 2) and coercivity spectra (Fig. 3), we suggest that the principal carrier of component P is magnetite and of component S, haematite. The mean remanence directions of the components are illustrated in Fig. 5. The remanence directions with corresponding virtual geomagnetic poles of different stages are given in Table 2. In the following, the palaeomagnetic results are discussed according to the stages.

The pyritizied sandstone samples from the lower Ordovician Tremadocian (Pakerort) Stage do not give any stable palaeomagnetic results. The argillite schists revealed some components but these are, from sample to sample, oriented chaotically. According to IRM experiments and AF demagnetization data, the rocks carry an intermediate-coercivity (removed between 0.3 and 0.6 T) and a high-coercivity mineral phases. The glauconitic sandstones of the Floian (Billingen) Stage reveal no stable components because the intensity of NRM decays under the detection limit within the very first AF demagnetization steps (coercivities less than 0.2 T). However, the glauconitic lime- and dolostones from Pakri and Mäekalda reveal an intermediate-coercivity (<0.6 T) SE-directed component P (Fig. 5) that in many specimens coexists with a low-coercivity component of likely viscous origin. Thermal demagnetization experiments reveal the component up to 300–350 °C, however the Curie temperature of the carrier cannot be identified due to mineralogical alterations in higher temperatures (Fig. 4b). According to the Lowrie test (Fig. 3e), magnetite is responsible for the intermediate-coercivity component. However, there exist other low-coercivity ferromagnetic minerals (pyrrhotite/maghaemite) with rather chaotically oriented remanence directions, responsible for the failure of the thermal treatment. In samples from Saka, some samples show a similar P component, however due to low number of specimens and high scatter between samples, they were not included into statistics. In Tõrvajõe, a stable high-coercivity and high-temperature component S, carried by haematite, has been identified in the glauconitic limestones of Billingen Stage. The same S component, carried by haematite (Figs 3d and e), was found also in the glauconitic lime- and dolostones of the Dapingian (Volkhov) Stage in rocks from NE Estonia (Sõtke, Tõrvajõe and Narva sites). The S component is strongly related to the haematite-bearing reddish patches as the grey Dapingian Stage carbonate rocks in the NW sites do not carry it. The Dapingian Stage glauconitic lime- and dolostones in Pakri, Mäekalda and Sõtke sites also carry component P (Figs 4c, d, g and h). Most of the unidentified directions at these sites are of low coercivity (removed at ≤0.4 T); the few high-coercivity components are directed SW to NW. In lime- and dolostones of Darriwilian (Kunda) Stage, only the component S with reversed polarity is identified (Figs 4e and f). It is represented in few samples from NE locations. The Lowrie test from the Narva site shows that haematite carries the high-coercivity component, however it also indicates the presence of low- and intermediate-coercivity mineral phases. Upper Darriwilian (Aseri) Stage limestones with iron (goethite) ooids also carry the component P. However, not goethite but magnetite is responsible for the palaeomagnetic signal. The goethite signal with high-coercivity and low Curie temperature (Fig. 3e) tends to be directed NE, a direction close to the present earth field (PEF). Chaotically oriented directions in low-coercivity phases are present in Aseri as well.

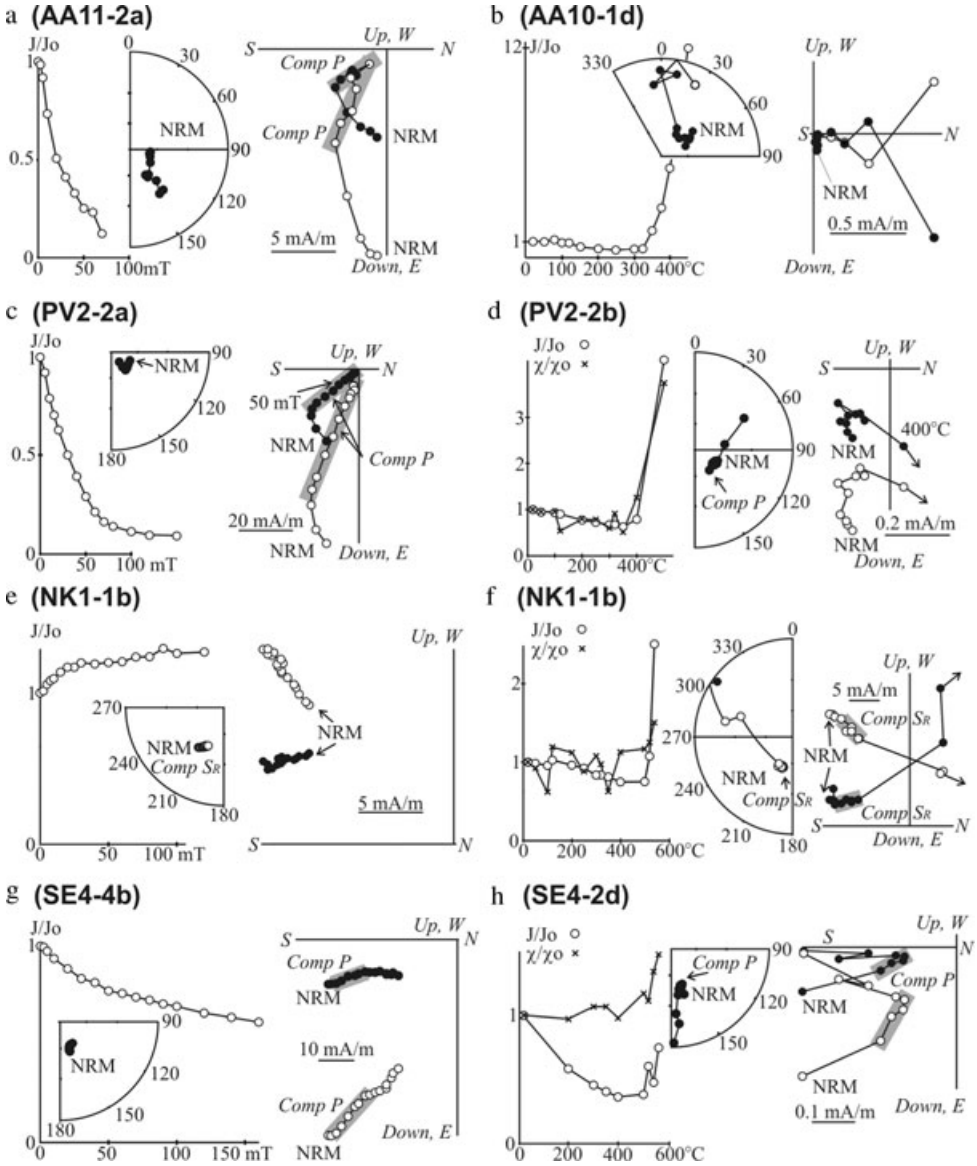


Figure 4. Examples of alternating field (a, c, e and g) and thermal (b, d, f and h) demagnetization behaviour of studied lithologies. Samples AA11 (a) and AA10 (b) Mäekalda, Billingen Stage; sample PV2 (c and d) Pakri, Volkhov Stage; sample NK1 (e and f) Narva, Kunda Stage; sample SE4 (g and h) Sõtke, Volkhov Stage (see Fig. 1 for sites and stages). (Left) Relative intensity and susceptibility (J/J_0 and χ/χ_0) decay curves, where the demagnetizing alternating field (mT) or temperature (°C) is given on the horizontal axis; (middle) stereographic projections of directional data on demagnetization; (right) orthogonal demagnetization diagrams, where open (closed) symbols denote vertical (horizontal planes). The thermal demagnetization of sample NK1 (f) was started after the treatment with alternating field (e).

Upper Darriwilian (Lasnamägi) Stage limestones reveal only low-coercivity components, but a stable S_R component was isolated in some samples (Table 1). The few samples from upper Darriwilian (Uhaku) Stage show, among low-coercivity and low-temperature components, a stable high-coercivity NE downward-directed S_N component.

4 DISCUSSION

To define ages for components P and S, their palaeomagnetic poles (Table 2) were plotted on the apparent polar wander path (APWP) for Baltica (Fig. 6; Torsvik & Cocks 2005) and compared with previously known poles. The comparison confirms early and middle Ordovician ages (488.3–460.9 Ma; Fig. 1) for the mean Floian (P_F: Lat = 25.0 °N, Long = 50.8 °E, $A_{95} = 7.2°$) and Dapingian to Darriwilian (P_{DD}: Lat = 11.4 °N, Long = 39.1 °E, $A_{95} = 6.7°$) poles, coinciding with the poles from the approximately same stratigraphic sequences in Scandinavia and St Petersburg area in Russia (Claesson 1978; Torsvik & Trench 1991a; Perroud *et al.* 1992; Torsvik *et al.* 1995; Smethurst *et al.* 1998; Fig. 6). A similar pole was previously obtained also by Mertanen (2006) from the Dapingian Stage glauconitic limestone in Pakri and from Aseri Stage Fe-ooid bearing limestone in Väo, near Tallinn. The P component is carried by magnetite that, most likely, was produced in locally slightly reducing (for the pE vs. pH equilibrium diagram of the Fe–S–H_2O system; see Henshaw & Merrill 1980) subbottom marine environment with an abundant supply of Fe (detrital three-valent iron mineral). The presence of glauconite in the sequences that carry the P component indicates low or negative sedimentation rate, locally reducing

© 2009 The Authors, *GJI*, **180,** 147–157
Journal compilation © 2009 RAS

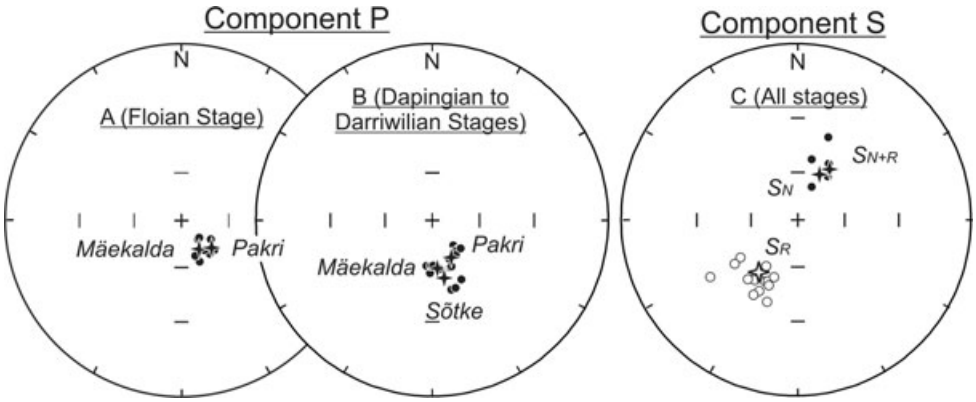


Figure 5. Equal-angle stereonet projections of *in situ* remanence directions of samples (dots) and corresponding mean directions (stars). Open (closed) symbol denotes upward(downward)-pointing remanence. Mean values for different sites, corresponding to stages, are showed for component P (see Table 2 for reference directions and corresponding virtual geomagnetic pole positions).

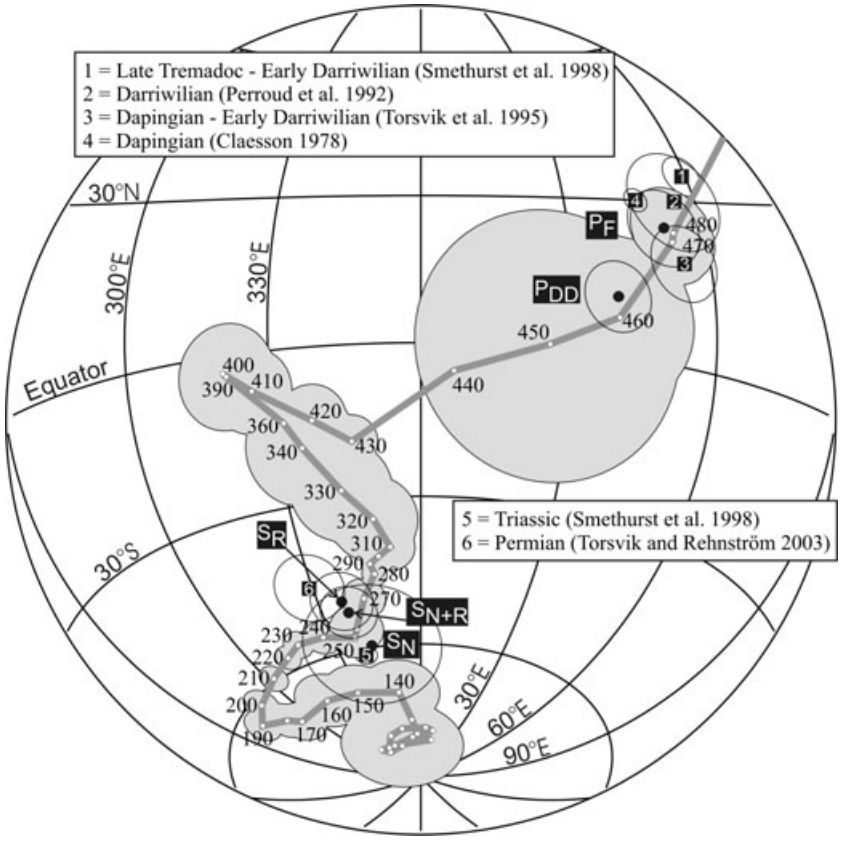


Figure 6. Combination of apparent polar wander path (thick grey line) for Baltica with running mean poles (Ma) and A_{95} confidence areas after Torsvik & Cocks (2005). Poles of this study (Table 2) with their A_{95} confidence areas are shown with black dots (P = palaeomagnetic pole of the primary component; S = palaeomagnetic pole of the secondary component; subindexes DD, F, N and R denote Dapingian to Darriwilian stages, Floian Stage, normal and reversed polarities, respectively). The black numbered squares represent palaeomagnetic poles from previous studies (see text for details).

conditions in a generally oxidizing environment and an abundant supply of K and Fe (Hower 1961; Kuzmann *et al.* 2008). Magnetite could also link to early dolomitization (e.g. Elmore *et al.* 1994). Anyway, the P component is likely not depositional but has been formed in a chemical process.

The individual poles calculated as site means for three successive stages (Floian, Dapingian, Darriwilian; Table 2) align approximately along the Ordovician sequence of the APWP of Baltica (Fig. 6). However, as the Dapingian Stage poles from different sites are more scattered along the APWP, the age of the mean pole is a few millions years too young. Because of low accumulation rate, we rule out the effect of unaveraged secular variation, as samples and even specimens contain material that has accumulated and suffered diagenesis within thousands of years. Further, as magnetite is likely of diagenetic in origin, the inclination error cannot be a reason for the scatter. Assuming the APWP is reliable, it is most likely due to imperfect separation of the component P and relatively low number of sites and samples. Lack of the dual polarity in P-poles suggests that our data belong to the long-lasting reversed polarity epoch prevailing in Early and Middle Ordovician epochs

(e.g. Torsvik *et al.* 1995). Comparison of magnetostratigraphic scheme from the Gullhøgen Quarry (Torsvik *et al.* 1995) with our sampled intervals gives polarity zone R1 for Billingen and Volkhov stages samples (Floian and Dapingian stages) and R4 for Aseri Stage samples (Darriwilian Stage).

Pole S (Lat = 52.2 °N, Long = 157.9 °E; A_{95} = 5.3°) plots on the younger part (250–270 Ma) of the APWP and gives middle to late Permian age. Dual polarity of the pole hints a period after the ∼50 million years Carboniferous and early Permian Reversed Polarity Superchron that was first terminated in the middle Permian (e.g. Khramov 1987; Szurlies *et al.* 2003), approximately in the middle to upper part of the Wordian Stage (268.0–265.8 Ma). Rest of the middle to upper Permian exhibit two prominent (Capitan N and Chang N) and shorter normal-polarity intervals (Steiner 2006). As our normal-polarity haematite-carried pole S_N combines with a time interval of 270–230 Ma in the Baltica's APWP (the closest is virtual pole at ∼250 Ma), we suggest it has been formed during one (or some) of the above-described normal-polarity intervals. The secondary remagnetization component resides in (titano)haematite as illustrated by mineralogical and confirmed by rock magnetic studies. The mineralogical relationship of the dolomite with haematite in the reddish dolostones (mainly in NE Estonia) suggests that the Fe–(Ti-)oxide post-dates the diagenetic dolomitization (Fig. 2e). Breakdown of glauconite may have been one source for haematite, but the microscopic studies hint to oxidation of sulphides and possible dedolomitization (e.g. Elmore *et al.* 1985) as well. Haematite concentrations in the Estonian sedimentary sequence have local occurrences whereas their frequency depends on region. That hints to local processes and local source of iron. However, similar Carboniferous to Triassic remagnetizations are reported from almost all the studied lower Palaeozoic outcrops in Fennoscandia and NE Russia (e.g. Perroud *et al.* 1992; Smethurst *et al.* 1998; Torsvik & Rehnström 2003). The authors have attributed the component to different processes. Smethurst *et al.* (1998) showed haematite and goethite as carriers of the Triassic component and suggested that the remanence represents chemical remanent magnetization due to breakdown of glauconite in the near-surface environment. Torsvik & Rehnström (2003) tied the Permian overprint in Komstad limestones of Bornholm with cooling/uplift (Hansen 1995). In Ordovician sequence of St Petersburg area, the component is interpreted to be related to the tectonic events at the Urals and in Western Europe (Lubnina 2004). Also the crystalline Svecofennian basement in the shear zones of southern Finland record a similar remanence component as the S component in Estonia. Mertanen *et al.* (2008) described two Phanerozoic events that date to (i) Silurian to Devonian and (ii) Permian to mid-Triassic, and tied the older event with tectonic reactivation coupled with following sedimentation after the Caledonian Orogeny. The origin of the younger component was not resolved, but it was connected to some late sedimentary processes after the Caledonian Orogeny.

Worldwide, there exist hundreds of documented indications of a late Palaeozoic secondary magnetization in different rock types. Events that have had a direct or oblique effect to the formation of a Permian remagnetization have usually been related to some local or regional processes, for example Alleghenian Orogeny in North America (Miller & Kent 1989; Beaubouef *et al.* 1990; Lewchuk *et al.* 2003; Cederquist *et al.* 2006) and Africa, Hunter-Bowen Orogeny in Australia (Lackie & Schmidt 1993; Klootwijk 2003), or magmatic activity in Siberia (Gallet & Pavlov 1996; Kravchinsky *et al.* 2002) and China (Yianping *et al.* 1990). The processes fit into the global frame in general—to the formation and evolution of the Pangaea supercontinent.

After the closure of the Iapetus Ocean (formation of the Laurasia continent) in Silurian, the present Baltic Sea region remained in the centre of the continent at the latitude of about 30 °N, in warm and semiarid climate. Permian widespread regression of the seas (Zharkov & Chumakov 2001, and references therein) coupled with the presence of haematite hints that the base level of erosion lowered and upper sediments were denudated when the oxidizing fluids could reach the older sedimentary sequences. As the sequence was never buried deeply nor heated (maximum burial depth for the underlying lower Cambrian clay in northern Estonia did not exceed 800–1000 m and a maximum temperature of ∼35 °C, assuming a constant thermal gradient of 20 °C km^{-1}; Kirsimäe *et al.* 1999; Kirsimäe & Jørgensen 2000), the termoviscous remagnetization due to cooling from elevated burial temperatures (e.g. Kent 1985) is ruled out. Therefore, it is interpreted that haematite that carries the component S is most likely of chemical origin and indicates near-surface alteration by meteoric fluids.

Our results specify the geological history of N Estonia in some details. Deposition of limestones in a shallow NE part of the Ordovician-Silurian Baltic Basin (Nestor & Einasto 1997) was followed by diagenetic dolomitization of limestones in some parts of the basin (Kiipli 1983). The primary detrital iron oxides were likely sulphurated due to anoxic conditions during diagenesis and following marine sedimentation at *Late Ordovician* to *late Silurian*. Early diagenetic magnetite was partly formed in those processes. During *late Silurian to Early Devonian*, a far-field expression of the compression by Caledonian tectonics, resulting to formation of mainly NE-SW-oriented faults, uplift, erosion and weathering of some of the covering Silurian and Ordovician deposits (Puura *et al.* 1999) took place. Because of different distance from the Caledonian front, the tectonic uplift and corresponding depth of erosion differed between NW and NE Estonia as the eastern part is much more eroded. During *Middle to Late Devonian*, the eroded slightly southerly dipping sequences were buried under the transgressive Devonian deposits, which were probably some hundreds of metres thick. Because of the above-mentioned differences in rates of uplift and erosion, base for the Devonian sediments was composed of ∼200-m-thick Ordovician to lower Silurian carbonate suite in NW Estonia but only 20- to 30-m-thick Middle to Lower Ordovician sequence in NE Estonia. Palaeogeological conditions similar to NE Estonia existed in the wide region until the Lake Ladoga area (Puura *et al.* 1999). The predominantly continental period lasted from *Carboniferous to Palaeogene.* In middle and late Permian, haematite, which according to the present study post-dates dolomites, has sporadically been precipitated into the pore space of dolostones. During the *Neogene* continental period, the Ordovician deposits became exposed near/or at the klint of N Estonia to their present position and haematite, as the end-member of the oxidation cycle, remained invariable.

5 CONCLUSIONS

Two components of remanent magnetization were identified in the Floian to Darriwilian carbonate sequence of N Estonia. A southeasterly downward-directed component (P) dates to Ordovician, and a south-westerly upward-directed component with a reversed counterpart, component (S) represents Permian remagnetization. Mineralogical studies revealed magnetite responsible for the mid-coercivity component P whereas the carrier of the high-coercivity, high-unblocking-temperature component S is haematite. The presence of goethite and maghaemite has been indicated by

© 2009 The Authors, *GJI*, **180,** 147–157
Journal compilation © 2009 RAS

magnetic tests as well, but they do not carry stable remanent magnetizations.

Component P residing in magnetite has been identified in sequences that contain abundant grains of glauconite (Billingen, Volkhov and Aseri regional stages). The magnetite formed in early diagenesis by chemical processes in locally slightly reducing sub-bottom marine environment with an abundant supply of Fe and low or negative sedimentation rate. Component S, residing in haematite, was identified in the reddish dolostones in NE Estonia. The red impregnation that occurs independent from stratigraphic layering is of chemical origin and most likely indicates near-surface alteration by meteoric fluids, coupled with mid-continent position of the studied area and Permian widespread regression of the seas. The pre-mid-Devonian deep erosional truncation of Ordovician deposits was probably responsible for development of preconditions for penetration of meteoric fluids from Devonian deposits into the lower Ordovician limestones, and therefore for the chemical formation of haematite with the mid to late Permian palaeomagnetic signatures.

ACKNOWLEDGMENTS

Authors thank Kalle Suuroja, Sten Suuroja, Tõnis Saadre (Geological Survey of Estonia), Tiiu Elbra (Helsinki University) and Matti Leino (Geological Survey of Finland) for their assistance in the field and laboratory. Authors also thanks the reviewers, Richard Elmore and Trond Torsvik, and an editor Cor Langereis for critical comments and corrections. The Estonian Science Foundation (grant nos. 5500 and 6613) supported the study.

REFERENCES

Beaubouef, R., Casey, J.F., Hall, S.A. & Evans, J., 1990. A paleomagnetic investigation of the Lower Ordovician St. George Group, Port-au-Port Peninsula, Newfoundland: implications for the Iapetus Ocean and evidence for Late Paleozoic remagnetization, *Tectonophysics,* **182,** 337–356.

Butler, R.F., 1992. *Paleomagnetism: Magnetic Domains to Geologic Terranes,* Blackwell Scientific Publications, Oxford.

Cederquist, D.P., Van Der Voo, R. & Pluijm, B.A., 2006. Syn-folding remagnetization of Cambro-Ordovician carbonates from the Pennsylvania Salient post-dates oroclinal rotation, *Tectonophysics,* **422,** 41–54.

Claesson, C., 1978. Swedish Ordovician limestones: problems in clarifying their directions of magnetizations, *Phys. Earth planet. Inter.,* **16,** 65–72.

Cocks, L.R.M. & Torsvik, T.H., 2005. Baltica from the late Precambrian to mid-Palaeozoic times: the gain and loss of a terrane's identity, *Earth-Sci. Rev.,* **72,** 39–66.

Einasto, R. & Mens, K., 1996. Geology. Basement (in Estonian), in *Pakri Peninsula–Nature and Human Activity,* pp. 16–23, ed. Kink, H., Estonian Academy Publishers, Tallinn.

Elmore, R.D., Dunn, W. & Peck, C., 1985. Absolute dating of dedolomitization by means of paleomagnetic techniques, *Geology,* **13,** 558–564.

Elmore, R.D., Cates, K., Gao, G. & Land, L., 1994. Geochemical constraints on the origin of secondary magnetizations in the Cambro-Ordovician Royer Dolomite, Arbuckle Mountains, southern Oklahoma, *Phys. Earth planet. Inter.,* **85,** 3–13.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc., Lond. A.,* **217,** 293–305.

Gallet, Y. & Pavlov, V., 1996. Magnetostratigraphy of the Moyero river section (northwestern Siberia): constraints on geomagnetic reversal frequency during the Early Paleozoic, *Geophys. J. Int.,* **125,** 95–105.

Hansen, K., 1995. Fennoscandian borderzone: thermal and tectonic history of a tuffaceous sandstone and granite from fission track analysis, Bornholm, Denmark, *Tectonophysics,* **244,** 153–160.

Hartz, E.H. & Torsvik, T.H., 2002. Baltica upside down: a new plate tectonic model for Rodinia and the Iapetus Ocean, *Geology,* **30,** 255–258.

Henshaw, P.C. & Merrill, R.T., 1980. Magnetic and chemical change in marine sediments. *Rev. Geophys. Space Phys.,* **18,** 483–504.

Hower, J., 1961. Some factors concerning the nature and origin of glauconite, *Am. Mineral.,* **46,** 313–333.

Jõeleht, A. & Kukkonen, I.T., 2002. Physical properties of Vendian to Devonian sedimentary rocks in Estonia, *GFF,* **124,** 65–72.

Kaljo, D. *et al*., 1986. The Cambrian-Ordovician boundary in the Baltic-Ladoga clint area (North Estonia and Leningrad region, USSR), *Proc. Acad. Sci. Eston. SSR, Geol.,* **35,** 97–108.

Kent, D.V., 1985. Thermoviscous remagnetization in some Appalachian limestones, *Geophys. Res. Lett.,* **12,** 805–808.

Khramov, A.N., 1987. *Paleomagnetology,* Springer-Verlag, Berlin.

Kiipli, T., 1983. On the genesis of Ordovician and Silurian dolomites at the contact with Devonian deposits (in Russian), *Proc. Acad. Sci. Eston. SSR, Geol.,* **32,** 110–117.

Kirsimäe, K. & Jorgensen, P., 2000. Mineralogical and Rb-Sr isotope studies of low-temperature diagenesis of Lower Cambrian clays of the Baltic paleobasin of North Estonia, *Clays Clay Miner.,* **48,** 95–105.

Kirsimäe, K., Jorgensen, P. & Kalm, V., 1999. Low-temperature diagenetic illite-smectite in Lower Cambrian clays in North Estonia, *Clay Miner.,* **34,** 151–163.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Klootwijk, C.T., 2003. Carboniferous paleomagnetism of the Werrie Block, northwestern Tamworth Belt, and the New England pole path, *Aust. J. Earth Sci.,* **50,** 865–902.

Kravchinsky, V.A., Konstantinov, K.M., Courtillot, V., Savrasov, J.J., Valet, J.-P., Cherniy, S.D., Mishenin, S.G. & Parasotka, B.S., 2002. Paleomagnetism of East Siberian traps and kimberlites: two new poles and paleogeographic reconstructions at about 360 and 250 Ma, *Geophys. J. Int.,* **148,** 1–33.

Kuzmann, E., Weiszburg, T.G., Tóth, E. & Garg, V.K., 2008. Mössbauer characteristics of glauconitisation, *Hyperfine Interact.,* **186,** 1–8.

Lackie, M.A. & Schmidt, P.W., 1993. Remagnetization of Strata during the Hunter-Bowen Orogeny, *Explor. Geophys.,* **24,** 269–274.

Lewchuk, T.M., Evans, M. & Elmore, D., 2003. Synfolding remagnetization and deformation: results from Paleozoic sedimentary rocks in West Virginia, *Geophys. J. Int.,* **152,** 266–279.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.,* **17,** 159–162.

Lubnina, N., 2004. Paleomagnetic investigations of the Ordovician rocks from St. Petersburg area: age of remagnetizations and their correlation with tectonic events. in *5th Nordic Paleomagnetic Workshop. Extended abstracts,* pp. 103–107, ed. Mertanen, S., Geological Survey of Finland, Espoo.

Miller, J.D. & Kent, D.V., 1989. Paleomagnetism of the Upper Ordovician Juniata Formation of the Central Appalachians revisited again, *J. geophys. Res.,* **94,** 1843–1849.

Mägi, S., 1984. A characterization of the type section of the Ontika subseries (in Russian), *Proc. Acad. Sci. Eston. SSR, Geol.,* **33,** 104–112.

Männil, R. & Meidla, T., 1994. The Ordovician System of East European Platform, *IUGS Publ.,* **28,** 1–52.

McFadden, P.L. & McElhinney, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

Mens, K., Viira, V., Paalits, I. & Puura, I., 1989. Cambrian-Ordovician boundary beds at Mäekalda, Tallinn, north Estonia, *Proc. Acad. Sci. Eston. SSR, Geol.,* **38,** 101–111.

Mertanen, S., 2006. *Paleomagnetic and rock magnetic studies from Middle Ordovician limestones in Väo and Pakri, northern Estonia,* Geological Survey of Finland Report.

Mertanen, S., Airo, M.-L., Elminen, T., Niemelä, R., Pajunen, M., Wasenius, P. & Wennerström, M., 2008. Paleomagnetic evidence for Mesoproterozoic–Paleozoic reactivation of the Paleoproterozoic crust in southern Finland, *Geological Survey of Finland, Special Paper,* **47,** 215–252.

Nestor, H. & Einasto, R., 1997. Ordovician and Silurian carbonate sedimentation basin, in *Geology and Mineral Resources of Estonia,* pp. 192–204, eds Raukas, A. & Teedumäe, A., Estonian Academy Publishers, Tallinn.

Orviku, K., 1940. Lithologie der Tallinna-Serie (Ordovizium, Estland) I, *Publications of the Geological Institution of the University of Tartu,* **58,** 1–249.

Orviku, K.K., 1960. About the lithostratigraphy of the Volkhov and Kunda Stages in Estonia (in Russian), *Studies of the Institute of Geology at the Academy of Sciences of the Estonian SSR,* **5,** 45–87.

Perroud, H., Robardet, M. & Bruton, D.L., 1992. Palaeomagnetic constraints upon the palaeogeographic position of the Baltic Shield in Ordovician, *Tectonophysics,* **201,** 97–120.

Popov, V., Iosifidi, A., Khramov, A., Tait, J. & Bachtadse, V., 2002. Paleomagnetism of the Upper Vendian sediments from the Winter Coast, White Sea region, Russia: implications for the palaeogeography of Baltica during Neoproterozoic times, *J. geophys. Res.,* **107,** doi:10.1029/2001JB001607.

Puura, V., Vaher, R. & Tuuling, I., 1999. Pre-Devonian landscape of the Baltic Oil-Shale Basin, NW of the Russian platform, in *Uplift, Erosion and Stability: Perspectives on Long-term Landscape Development,* Vol. 162, pp. 75–83, eds Smith, B.J. *et al*., Geological Society of London, Special Publication.

Smethurst, M.A., Khramov, A.N. & Pisarevsky, S., 1998. Palaeomagnetism of the Lower Ordovician Orthoceras Limestone, St. Petersburg, and a revised drift history for Baltica in the early Palaeozoic, *Geophys. J. Int.,* **133,** 44–56.

Steiner, M.B., 2006. The magnetic polarity time scale across the Permian-Triassic boundary, in *Non-Marine Permian Biostratigraphy and Biochronology,* Vol. 265, pp. 15–38, eds Lucas, S.G., Cassinis, G. & Schneider, J.W., Geological Society of London, Special Publication.

Szurlies, M.B., Gerhard, H., Menning, M., Nowaczyk, N.R. & Kaeding, K.C., 2003. Magnetostratigraphy and high-resolution lithostratigraphy of the Permian-Triassic boundary interval in central Germany, *Earth planet. Sci. Lett.,* **212,** 263–278.

Teedumäe, A., Kallaste, T. & Kiipli, T., 2001. Aspects of the dolomitization of the Mõhküla beds (Silurian, Estonia), *Proc. Eston. Acad. Sci.,* **50,** 190–205.

Teedumäe, A., Shogenova, A. & Kallaste, T., 2006. Dolomitization and sedimentary cyclicity of the Ordovician, Silurian, and Devonian rocks in South Estonia, *Proc. Eston. Acad. Sci.,* **55,** 67–87.

Torsvik, T.H. & Cocks, L.R.M., 2005. Norway in space and time: a Centennial cavalcade, *Norwegian J. Geol.,* **85,** 73–86.

Torsvik, T.H. & Rehnström, E.F., 2001. Cambrian palaeomagnetic data from Baltica: implications for true polar wander and Cambrian palaeogeography, *J. geol. Soc. Lond.,* **158,** 321–329.

Torsvik, T.H. & Rehnström, E.F., 2003. The Tornquist Sea and Baltica-Avalonia docking, *Tectonophysics,* **362,** 67–82.

Torsvik, T.H. & Trench, A., 1991a. The Lower-Middle Ordovician of Scandinavia: southern Sweden "revisited", *Phys. Earth planet. Inter.,* **65,** 283–291.

Torsvik, T.H. & Trench, A., 1991b. Ordovician magnetostratigraphy: Llanvirn-Caradoc limestones of the Baltic Platform, *Geophys. J. Int.,* **107,** 171–184.

Torsvik, T.H., Trench, A., Lohmann, K.C. & Dunn, S., 1995. Lower Ordovician reversal asymmetry—an artefact of remagnetization or nondipole field disturbance, *J. Geophys. Res.—Solid Earth,* **100,** 17 885–17 898.

Vaher, R.M., Puura, V.A. & Erisalu, E.K., 1962. *Tectonism of NE Estonia (in Russian), in Geology of Paleozoic,* Valgus, Tallinn.

Viira, V., Löfgren, A., Mägi, S. & Wickström, J., 2001. An Early to Middle Ordovician succession of conodont faunas at Mäekalda, northern Estonia, *Geol. Mag.,* **138,** 699–718.

Viira, V., Mens, K. & Nemliher, J., 2006. Lower Ordovician Leetse Formation in the North Estonian Klint area. *Proc. Eston. Acad. Sci., Geol.,* **55,** 156–174.

Yianping, L. *et al*., 1990. A Devonian paleomagnetic pole from red beds of the Tarim Block China, *J. geophys. Res.,* **95,** 19 185–19 198.

Zharkov, M.A. & Chumakov, N.M., 2001. Paleogeography and sedimentation settings during Permian-Triassic reorganizations in biosphere, *Stratigr. Geol. Correl.,* **9,** 340–363.

Zijderveld, D.J., 1967. A.C. demagnetization in rocks: analysis of results, in *Methods in Paleomagnetism,* pp. 254–286, eds Collonson, D.W. *et al*., Elsevier, Amsterdam.

© 2009 The Authors, *GJI*, **180,** 147–157
Journal compilation © 2009 RAS